



Moisture diffusion in maize hot air drying based on multi-component structure

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ABSTRACT

Grain-drying is essentially a complicated process of heat and mass transfer, which involves the coupling between the grain-particle system and the drying medium. Maize kernels are geometrically intricate heterogeneous structures, with complex internal mechanisms governing moisture diffusion and heat transfer. The low-field nuclear magnetic resonance (LF-NMR) and magnetic resonance imaging (MRI) techniques were employed to explore the migration and diffusion characteristics of water in diverse bound forms within grain particles throughout the drying process. A three-dimensional, multi-component drying model, specifically tailored for maize kernels was developed, considering the heterogeneity of moisture diffusion coefficients among different components, and simulated the temperature and moisture variations during the maize kernel drying process. Four distinct forms of moisture were observed within maize kernels. During hot-air drying, irrespective of the drying temperature, the binding strength of each form of moisture ascends in tandem with the decline in the moisture ratio. The kernels remain in the falling-rate drying stage, with a bidirectional migration phenomenon occurring between chemically-bound water and other forms of water. The error between the multi-component models and the experimental values was significantly smaller than that of single-component models, with the maximum error being less than 4.96 %. The heterogeneity among components determines the non-uniform distribution of both temperature and moisture content within the maize kernel during the drying procedure.

1. Introduction

Food security is a critical strategic foundation for maintaining social stability, national economic development, and national security (Haque et al., 2022). During harvesting, kernels have a high initial moisture content. High-moisture kernels accelerate microbial growth and spoilage, which negatively impacts processing, transportation, and storage (Boateng, 2023; Nunes et al., 2022; Ziegler et al., 2021). Drying constitutes an essential post-harvest procedure; however, the intricate architecture of maize kernels combined with the porous network within grain piles often leads to substantial heterogeneity in temperature and moisture distribution during the drying process. (Liu et al., 2022, 2023, 2023). A grain pile is composed through the accumulation of cereal kernels. The internal transfer of heat and moisture therein involves interactions at the scales of the grain, pore, and pile, thereby making it a multi-scale, multi-field coupling transfer problem (Ajani et al., 2021; Defraeye, 2014; Ho et al., 2013; Thorpe, 2008). The drying of a single

grain kernel serves as the basis for investigating macroscopic and microscopic transport phenomena within grain piles and pore spaces (ElGamal et al., 2014; Xiao et al., 2008).

Initial studies on the thermal drying characteristics of kernels involved constructing models at the kernel scale to determine the effective diffusion coefficient of moisture within the grain (Ghosh et al., 2008; Hacıhafızoglu, Cihan, & Kahveci, 2008; Singh et al., 2004; Takhar et al., 2011). Chen et al. developed a model for the effective diffusion of moisture in different components of maize kernels, using an Arrhenius relationship to describe its dependence on temperature and moisture content (Chen et al., 2009). Wei et al. developed a three-dimensional geometric model for maize kernels, using a coupled thermal-moisture transfer and elastic stress model to simulate the temperature and humidity gradients inside the kernel, as well as the distribution of drying stresses under various drying temperatures (Wei et al., 2019, 2020, 2023).

Moisture diffusion during maize drying is a complex process. During

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the drying process, various forms of moisture undergo mutual migration and diffusion. Evaporative moisture moves out of the maize kernel through diffusion, a process that significantly affects drying characteristics (Hu et al., 2019; Jia et al., 2020; Jiang et al., 2021). Maize kernels have complex, heterogeneous geometries, and treating them as homogeneous bodies leads to discrepancies in comparison to reality, preventing the accurate representation of moisture diffusion and heat transfer mechanisms (Wang, Che, & Wan, 2021; Wang, Che, & Wan, 2021). Low-field nuclear magnetic resonance (LF-NMR), as an effective approach for determining the moisture distribution and forms of bound water, is capable of evaluating the mobility, content, and distribution of moisture in plants (Menon et al., 1987; Partanen et al., 2004; Sharp et al., 1978).

In this study, a comprehensive approach integrating experimental investigations and numerical simulations was adopted to comprehensively analyze the moisture diffusion characteristics and migration mechanisms within individual maize kernels during the drying process. LF-NMR and MRI technologies were used to investigate the variations in internal moisture forms and their corresponding contents in maize kernels under different drying temperatures. Furthermore, a three-dimensional, multi-component drying model for maize kernels was also developed at the particle scale, incorporating the heterogeneous moisture diffusion coefficients among different kernel components. The proposed model was then employed to simulate the dynamic changes of temperature and moisture during the drying process. To verify the reliability and accuracy of the simulation method, it was validated against the experimental drying data. Different forms of moisture within the kernels were identified, and the distribution and migration mechanisms of moisture were investigated based on both the experimental and simulated results.

2. The experimental materials and methodologies

2.1. Materials

The maize variety employed in the current experimental study was Zhengdan 958, obtained from a warehouse in Henan Province, China. The freshly harvested maize had a dry basis moisture content ranging from approximately 20 %–36 % (d.b.), while the maize stored in the warehouse underwent drying, achieving uniform moisture levels at a lower percentage (Liu et al., 2022, 2023). Prior to the experimental procedures, maize kernels were carefully selected according to strict criteria, including uniform size distribution, intact surface morphology without visible cracks or damage, and optimal cleanliness to ensure experimental consistency and reliability. Three samples were dried for 72 h at 103 ± 2 °C in an electric heating constant-temperature blast-drying box, yielding an initial dry-basis moisture content of approximately 13.98 ± 0.12 % (d.b.) (ASAE, 2017). To analyze moisture diffusion during drying, high-moisture maize samples were prepared by rewetting. A specific mass of maize kernels was weighed, and water was added based on the initial and target moisture content calculations to reach a final moisture content of 29 % (d.b.). Three groups of samples were conditioned accordingly, resulting in an average dry-basis moisture content of 28.41 ± 0.16 % (d.b.), as confirmed by standard oven-drying measurements. The prepared samples were carefully sealed in polyethylene bags and subsequently stored at 4 °C within a temperature- and humidity-controlled chamber. Prior to the drying test, the samples were equilibrated to room temperature 24 h before use.

2.2. Experimental apparatus

The drying process of maize kernels was conducted in a DHG-9140A desktop electric constant-temperature blast drying box (Shanghai Jinghong Experimental Equipment Co., Ltd.), with internal dimensions of 450 mm × 550 mm × 550 mm. The box features a bottom fan to enhance forced vertical airflow, and the temperature control range is 10–250 °C,

with an accuracy of 0.1 °C and an air temperature fluctuation within 1 °C. The transverse relaxation time changes during maize kernels drying were measured using an NMI20-060H-I nuclear magnetic resonance imaging analyzer, which was produced by Suzhou Niumai Analytical Instruments Co., Ltd. The temperature of the permanent magnet within the analyzer was precisely maintained at 32.0 ± 0.01 °C. The magnetic field strength was set at 0.5 ± 0.06 T, and the diameter of the internal probe coil was 60 mm. MRI measurements were performed using the same analyzer, and grayscale images were processed using Niumai NMR image processing software to produce pseudo-color images. Fig. 1 shows the maize-kernel-drying device and experimental process flow. The drying operated in an open-cycle mode, where ambient air was continuously heated and circulated without humidity regulation. the relative humidity remained low and stable throughout the drying process.

2.3. Experimental procedure

Twenty intact and similarly sized maize kernels were selected from the prepared samples, and their initial mass was recorded. The kernels were carefully and uniformly placed at the base of a glass tube. Subsequently, the glass tube was positioned within the magnetic field region of the MRI analyzer. The transverse relaxation time (T_2) spectra of the maize kernels were measured using a CPMG pulse sequence. In the context of this sequence, the magnetic frequency was set to 17 MHz. The pulse durations for the 90° and 180° pulses were 8.0 and 17.0 μs, respectively. The number of echoes was 2000, the echo time was 184 μs, and a total of 16 cumulative scans were carried out. Upon completion, the maize kernels were removed and spread evenly in a single layer on a stainless steel sieve, ensuring no contact between kernels. The sieve was placed into the drying box, with the drying temperature set to 80 °C. The total drying time was 180 min, with samples removed every 30 min to record instantaneous mass and T_2 characteristic spectra. During this period, two kernels from the drying samples were placed separately into a 60 mm diameter tube for MRI scanning, capturing the moisture changes throughout the drying process. After 180 min, the drying box temperature was raised to 105 °C, and the kernels were dried for an additional 24 h to measure their absolute dry mass (AOAC, 2005). Samples corresponding to three drying temperatures, namely 60, 70, and 80 °C, were chosen for LF-NMR relaxation measurements and MRI. Additional moisture content measurements were conducted at 15-min intervals to match the simulation time steps and improve the accuracy of validation analysis.

3. The proposed numerical models and the simulation methods

3.1. The governing equations

The assumptions made for the solution of the heat-moisture-transfer equations during the drying process of maize kernels are as follows: (1) The initial temperature and moisture content within the maize kernels are uniformly distributed; (2) The shrinkage phenomenon of maize kernels during the drying process is neglected. (3) Moisture within the maize kernels diffuses in either liquid or gaseous form to the surface, where it undergoes vaporization or dissipation. (4) Throughout the entire drying process, the temperature, relative humidity, and airflow velocity in the vicinity of the kernels remain constant (Li, Yang, et al., 2023).

According to Fourier's law of heat conduction and the first law of thermodynamics, the heat Q_H accumulated or released by an infinitesimal body due to temperature changes and material variations can be expressed as:

$$Q_H = \frac{\partial}{\partial t} (\rho H) dx dy dz = (Q_x + Q_y + Q_z) + Q_t \quad (1)$$

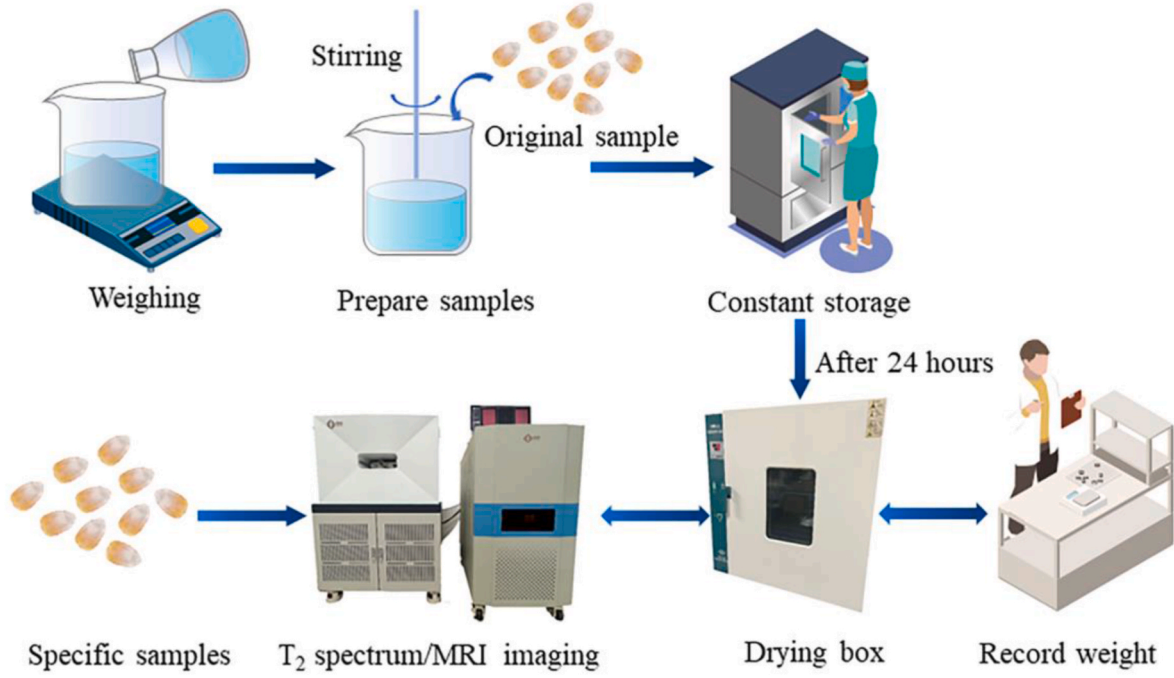


Fig. 1. Maize kernel drying device and experimental process.

where t is time, s; ρ represents the density of the maize kernel, kg/m^3 ; H denotes the latent heat of vaporization of water; (x, y, z) denote the three-dimensional Cartesian coordinates; Q_i ($i = x, y, z$) represents the heat generated by an arbitrary infinitesimal body at a specific location; and Q_t is the heat an produced by an infinitesimal body at any given time.

$$Q_x = \frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) dx dy dz \quad Q_y = \frac{\partial}{\partial y} \left(\lambda \frac{\partial T}{\partial y} \right) dx dy dz \quad Q_z = \frac{\partial}{\partial z} \left(\lambda \frac{\partial T}{\partial z} \right) dx dy dz \quad (2)$$

$$Q_t = q_v dx dy dz \quad (3)$$

where λ denotes thermal conductivity of the maize kernel, $\text{W/(m}\cdot\text{K)}$; T represents the temperature of the maize kernel, K; and q_v signifies the internal heat generation rate, $\text{J/(m}^3\cdot\text{s)}$.

Substituting the expressions of Q_i and Q_t to Q_H , the heat transfer within the maize kernels can be described as:

$$\frac{\partial}{\partial t} (\rho H) dx dy dz = \frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) dx dy dz + \frac{\partial}{\partial y} \left(\lambda \frac{\partial T}{\partial y} \right) dx dy dz + \frac{\partial}{\partial z} \left(\lambda \frac{\partial T}{\partial z} \right) dx dy dz + q_v dx dy dz \quad (4)$$

During the drying process, the internal moisture within the maize kernel evaporates. By differentiating the moisture - associated variable with respect to time, the following equation can be obtained, which is expressed as:

$$H = h_{fg} M + c_p T \quad \frac{\partial H}{\partial \tau} = c_p \frac{\partial T}{\partial \tau} - h_{fg} \frac{\partial M}{\partial \tau} \quad (5)$$

where h_{fg} represents the latent heat of vaporization of moisture within the maize kernel, J/kg ; M denotes the dry-basis moisture content, % (d. b.); c_p corresponds to the specific heat capacity of the maize kernel, $\text{J/(kg}\cdot\text{K)}$; and τ represents the characteristic time constant.

Given that moisture within the maize kernel is assumed to diffuse to its surface in either liquid or gaseous form before vaporizing or dissipating, the three-dimensional heat transfer mathematical model for the maize-kernel-drying process is expressed as follows (Li, Yang, et al., 2023):

$$\frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(\lambda \frac{\partial T}{\partial z} \right) + q_v = \rho c_p \frac{\partial T}{\partial \tau} - \rho h_{fg} \frac{\partial M}{\partial \tau} \quad (6)$$

The initial conditions and boundary conditions for the heat transfer process are:

$$T|_{t=0} = T_0 - \lambda \left(\frac{\partial T}{\partial x} + \frac{\partial T}{\partial y} + \frac{\partial T}{\partial z} \right) = h_t (T - T_a) \quad (7)$$

where T_0 is the initial temperature of the maize, K; h_t is the convective heat transfer coefficient, $\text{W/(m}^2\cdot\text{K)}$; and T_a represent the hot air drying temperature, K.

By analogizing to the external fluid passing through a sphere, the convective heat-transfer coefficient during the drying process of maize kernels can be calculated as follows (Li, Yang, et al., 2023):

$$\begin{cases} Nu = \frac{h_t d}{\lambda_a} = 2 + 0.552 Re^{0.53} Pr^{0.33} \\ Re = \frac{\nu_a \rho_a d}{\mu_a} \\ Pr = \frac{\mu_a c_a}{\lambda_a} \end{cases} \quad (8)$$

where Nu denotes the Nusselt number; d is the equivalent diameter of the maize kernel, m; λ_a represents the thermal conductivity of the air, $\text{W/(m}\cdot\text{K)}$; Re is the Reynolds number; Pr represents the Prandtl number; ν_a is the velocity flowing through the maize kernel, m/s ; ρ_a represents the density of the air, kg/m^3 ; μ_a is the air dynamic viscosity, $\text{Pa}\cdot\text{s}$; and c_a is the specific heat capacities of the air, $\text{J/(kg}\cdot\text{K)}$.

The differential equation that governs the moisture diffusion within the maize kernel is modeled in accordance with Fick's law. Based on the principle of mass conservation, the three-dimensional mass transfer mathematical model for the drying process of maize kernels is expressed as (Ostaneck & Iteleji, 2020):

$$\frac{\partial M}{\partial t} = D_{eff} \left(\frac{\partial^2 M}{\partial x^2} + \frac{\partial^2 M}{\partial y^2} + \frac{\partial^2 M}{\partial z^2} \right) \quad (9)$$

where D_{eff} is the effective moisture diffusion coefficient of the maize

kernel, m^2/s .

The initial and boundary conditions for the mass-transfer process are as follows:

$$M|_{t=0} = M_0 - D_{\text{eff}} \left(\frac{\partial M}{\partial x} + \frac{\partial M}{\partial y} + \frac{\partial M}{\partial z} \right) = h_m (M - M_e) \quad (10)$$

where M_0 represents the initial dry-basis moisture content of the maize kernel, % (d.b.); h_m denotes the convective mass-transfer coefficient, m/s ; and M_e is the equilibrium moisture content of the maize kernel, %.

By analogizing to the external fluid passing through a sphere, the convective mass-transfer coefficient during the drying process of maize kernels can be calculated as follows (wei et al., 2021):

$$\begin{cases} Sh = \frac{h_m d}{D_a} = 2 + 0.552 Re^{0.53} Sc^{0.33} \\ Sc = \frac{\mu_a}{\rho_a D_a} \end{cases} \quad (11)$$

where Sh is the Sherwood number; Sc is the Schmidt number; and D_a is the moisture diffusion coefficient in the drying medium, m^2/s .

3.2. The multi-component three-dimensional geometric model of the maize kernel

The geometric model of the maize kernel is established upon irregular shapes characterized by average dimensions. By extracting the boundary profiles and geometric parameters of the kernel's epidermis, hard endosperm, soft endosperm, and germ, the length, width, and height of the maize kernel could be determined as 11.84, 8.61, and 4.72 mm, respectively. A multi-component three-dimensional geometric model of the maize kernel was constructed using the SpaceClaim package of Ansys Fluent, incorporating a series of edge-smoothing treatments during the modeling process. Through the assembly and integration of all component shapes, both single-component and multi-component geometric models of the maize kernel were constructed, as depicted in Fig. 2.

3.3. The present mesh generation scheme

Considering the symmetrical characteristics of the maize kernel along its width axis, only half of the three-dimensional model was employed for the simulation. The middle cross-section was set as a symmetry boundary, and Ansys Fluent meshing was used to mesh the model, as shown in Fig. 3. The maximum mesh cell size was configured to 0.3 mm, with a mesh growth rate of 1.2. Regarding the half - model of the maize kernel, the total number of hexahedral elements was 20,625 in the single-component geometry and 59,284 in the multi-component geometry. Specifically, the number of hexahedral elements for the epidermis, hard endosperm, soft endosperm, and germ were 13,854, 26,543, 13,429, and 5,458, respectively. The geometric volume fractions of the different components within the maize kernel are shown in

Table 1.

3.4. The boundary condition settings and the solution methods

The commercial computational fluid dynamics (CFD) software Ansys Fluent, was employed to simulate the heat and moisture transfer during the drying of maize kernels. For all models, the initial temperature of the maize kernels was set at 25 °C, and the initial dry-basis moisture content was specified as 28.41 % (d.b.). The drying temperature was kept constant at 60, 70, and 80 °C, respectively. Table 2 provides the parameter settings for maize and air in the simulation (Wei et al., 2021). Non-slip boundary condition was applied to the kernel surface, and the transfer of temperature and moisture was simulated via the utilization of the coupled wall transfer method. A laminar flow model was adopted to simulate the convective process occurring between the kernels and air. The governing equations were resolved using the SIMPLE algorithm. The pressure term was discretized by means of the standard scheme. For all the remaining equations, a second-order upwind scheme was employed for discretization in order to mitigate numerical dissipation. To guarantee the accuracy of the solution, the time step was set to 1.0 s, and the simulation was terminated when the residuals of the monitored variables decreased to below 10^{-6} . The additional 50 steps further reinforced the reliability of our results while maintaining computational efficiency.

4. Results and discussion

4.1. Distribution of moisture inside maize kernels throughout the drying process

Fig. 4 illustrates the transverse relaxation time (T_2) inversion spectra obtained during the drying process of maize kernels at different temperatures (60 °C, 70 °C, and 80 °C). The T_{22} spectra revealed consistent patterns in the weakening of chemically bound water across all temperature conditions. As the initial moisture content decreased to the target level, a distinct leftward shift was observed in both the intensity peak of the T_{22} spectra and the transverse relaxation time (T_2). This shift indicates a significant reduction in the mobility of water molecules within the maize kernel matrix, reflecting the progressive immobilization of water during the drying process. The binding capacity of the various moisture forms increased during the drying process. Compared to the initial state, the average binding energy of T_{22} increased rapidly in the later stages (Ritota et al., 2008). The variations in T_2 peak positions corresponding to each drying stage at different temperatures were similar, suggesting that the binding force of different moisture forms was independent of temperature. In the T_2 spectra, the four peaks gathered more closely, with adjacent states of water undergoing mutual penetration and transfer. Throughout the drying process, the forms of moisture remained unchanged, though the moisture content of each form altered (Zhang et al., 2020). During drying, free water on the surface of maize kernels evaporated first, while bound water had to be

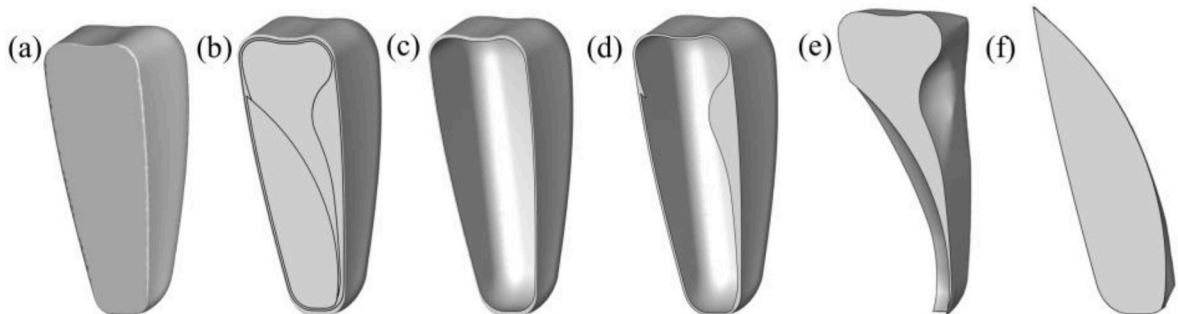


Fig. 2. Geometric models of maize kernels, (a) single-component, (b) multi-component, (c) pericarp, (d) hard endosperm, (e) soft endosperm, (f) germ.

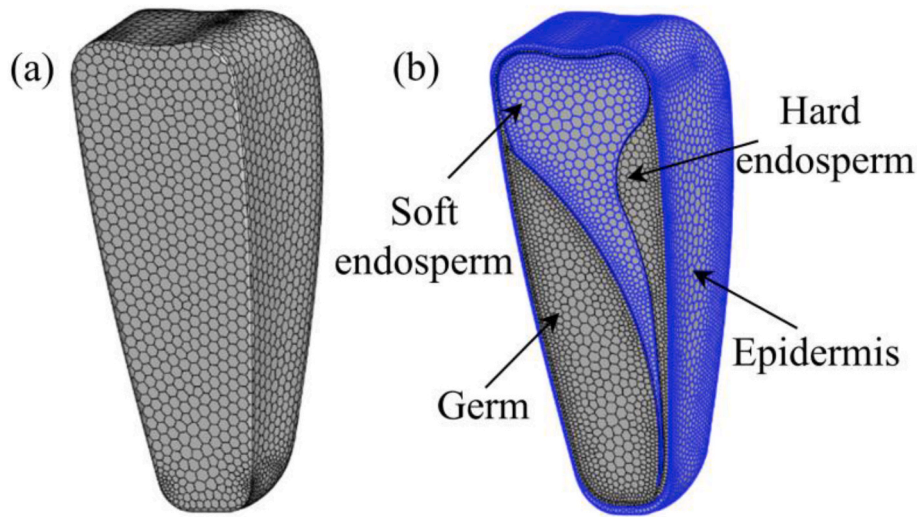


Fig. 3. Meshed geometry of the half-model of the maize kernel, (a) Single-component, (b) multi-component.

Table 1

Volume-related parameters of various components within the maize geometric model.

	maize kernel	epidermis	Hard endosperm	soft endosperm	germ
Volume/ mm ³	419.27	9.27	241.62	108.91	59.47
Content ratio	1	0.02	0.58	0.26	0.14

Table 2

Numerical simulation parameters for maize kernels and airflow characteristics.

Material	Simulation parameters	value
Maize	Kernel Density $\rho/(\text{kg}\cdot\text{m}^{-3})$	1240
	Specific heat $c_p/(\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1})$	$1470 + 3600M/(1 + M)$
	Thermal conductivity $\lambda/(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	$\exp(-1.74 - 3.7M + 4.72 \times 10^{-3} T + 6.48 M^3 - 1.5 \times 10^{-4} T^2 + 6.27 \times 10^{-2} MT)$
	Effective moisture diffusion coefficient of kernel $D_0/(\text{m}^2\cdot\text{s}^{-1})$	$7.817 \times 10^{-5} \exp(-4850/T + 5.5M)$
	Effective moisture diffusion coefficient of epidermis $D_1/(\text{m}^2\cdot\text{s}^{-1})$	$3.6 \times 10^{-6} \exp(-3559.1/T) M^{1.5}$
	Effective moisture diffusion coefficient of hard endosperm $D_2/(\text{m}^2\cdot\text{s}^{-1})$	$4.08 \times 10^{-6} \exp(-2506.6/T) M^{1.5}$
	Effective moisture diffusion coefficient of soft endosperm $D_3/(\text{m}^2\cdot\text{s}^{-1})$	$9.65 \times 10^{-7} \exp(-1588.1/T) M^{1.5}$
	Effective moisture diffusion coefficient of germ $D_4/(\text{m}^2\cdot\text{s}^{-1})$	$8.97 \times 10^{-6} \exp(-3020.2/T) M^{1.5}$
	Equilibrium moisture content M_e	$[\log(1 - RH_a)/(8.654 \times 10^5)/(T_a - 23.34)]^{1/1.8634}/100$
Air	Density $\rho_a/(\text{kg}\cdot\text{m}^{-3})$	$8.666 \times 10^{-6} T_a^2 - 4.318 \times 10^{-3} T_a + 1.288$
	Specific heat $c_a/(\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1})$	$4.834 \times 10^{-4} T_a^2 - 2.218 \times 10^{-2} T_a + 1007$
	Thermal conductivity $\lambda_a/(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	$2.401 \times 10^{-8} T_a^2 + 7.554 \times 10^{-5} T_a + 2.364 \times 10^{-2}$
	Dynamic viscosity $\mu_a/(\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1})$	$-3.238 \times 10^{-11} T_a^2 + 4.839 \times 10^{-8} T_a + 1.73 \times 10^{-5}$
	Diffusion coefficient of moisture vapor in air D_a	$3.229 \times 10^{-10} T_a^2 + 1.577 \times 10^{-7} T_a + 2.089 \times 10^{-5}$

transformed into free water before it could diffuse and be removed from the grain.

4.2. Migration of different forms of moisture during maize kernel drying

During the drying process, the relative change rate of each form of moisture is defined as the change in the peak area of a specific moisture form relative to its initial peak area. This relative change rate represents the dynamic characteristics of moisture migration. Fig. 5 presents the relative change rates (RT_{2i}) of various moisture forms during maize kernel drying at different temperatures. The RT_{2i} values, where positive and negative values indicate moisture content increase and decrease respectively, demonstrate distinct patterns across temperature conditions. Analysis revealed that the maximum RT_{2i} values occurred at specific moisture contents: 14.75 % 60 °C, 16.38 % for 70 °C, and 16.81 % for 80 °C drying temperatures. These critical points represent significant transitions in the moisture removal dynamics of different water forms within the kernel matrix. Higher drying temperatures were observed to enhance moisture migration efficiency. Notably, RT_{22} remained negative throughout the drying process, indicating a continuous decrease in T_{22} . As the temperature increased, the minimum value of the relative change rate shifted towards lower moisture content levels in the maize kernels. Specifically, RT_{21} dropped below zero at dry-base moisture contents of 16.79 %, 14.02 %, and 12.02 % for drying temperatures of 60 °C, 70 °C, and 80 °C, respectively. During this stage, T_{21} exhibited a progressive decrease as drying progressed, with alternating positive and negative RT_{21} values suggesting bidirectional migration between chemically bound water and other moisture forms.

4.3. Validation of moisture prediction during maize kernel drying

Fig. 6 compares the simulated and experimental moisture content curves for maize kernels at different temperatures during the drying process. The deviations between the simulated values and experimental values were significantly larger in the single-component model compared to the multi-component model. This discrepancy arises because the moisture diffusion coefficient in the single-component model represents an aggregate parameter, encompassing the diffusion abilities of all kernel components and influenced by their respective volume and mass ratio. As a result, the spatial variation in moisture diffusion across different parts of the maize kernel is not adequately captured. Specifically, the hard endosperm exhibits a relatively lower moisture diffusion capacity compared to the overall diffusion coefficient of the grain. When the hard endosperm constitutes a large volume

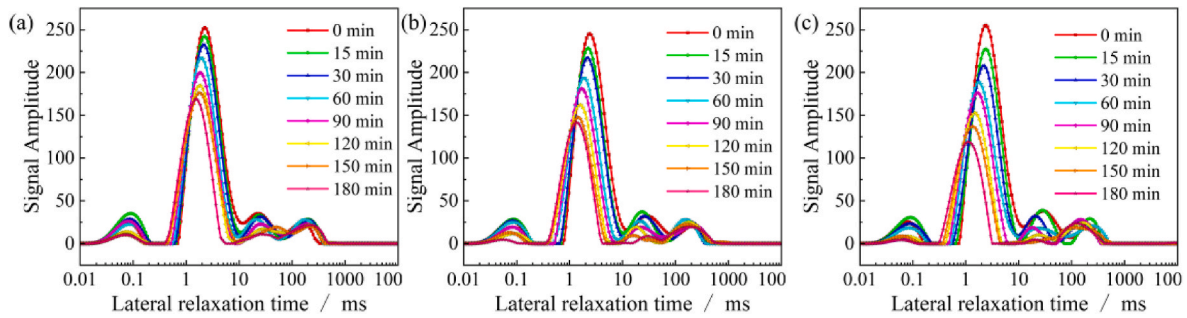


Fig. 4. The transverse relaxation time T2 spectrum of maize kernels during drying, (a) 60 °C, (b) 70 °C, (c) 80 °C.

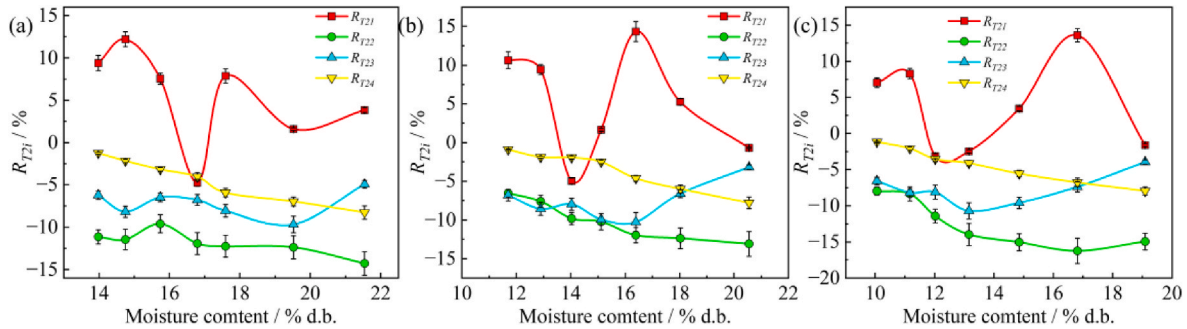


Fig. 5. Relationship between moisture changes in each phase in maize kernels during drying, (a) 60 °C, (b) 70 °C, (c) 80 °C.

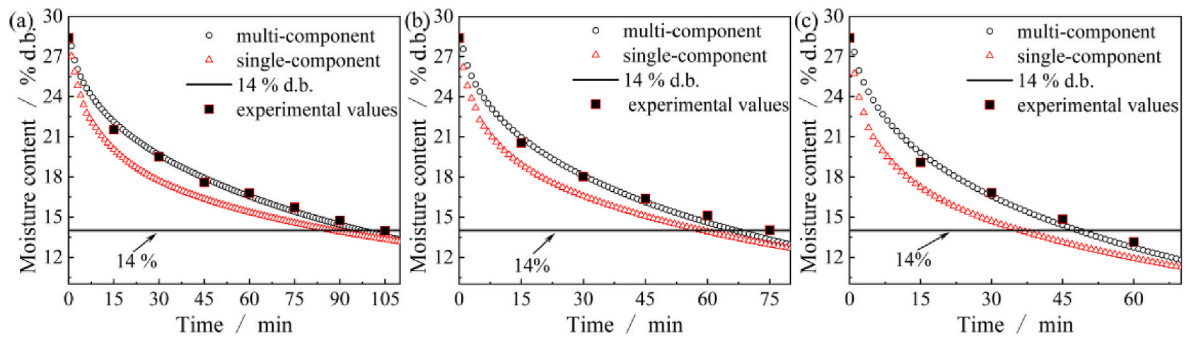


Fig. 6. The comparison of moisture simulation and experimental during maize kernel drying at different temperatures, (a) 60 °C, (b) 70 °C, (c) 80 °C.

fraction within the sample, it further reduces the measured moisture diffusion coefficient, leading to inaccuracies in the single-component model. In contrast, the multi-component model accounts for the heterogeneity of moisture diffusion across different kernel components, resulting in a significantly improved fit between simulated and experimental values.

4.4. Analysis of moisture changes during maize kernel drying

To investigate the moisture changes in maize kernels during drying, a temperature of 60 °C was selected. Assuming an initial uniform moisture distribution within the kernels, Fig. 7 illustrates the moisture distribution in the kernels at different times, as predicted by the single-component and multi-component models. As drying progressed, the moisture content decreased progressively from the surface to the interior. During the initial drying phase, the surface moisture rapidly approached equilibrium moisture content, while the internal moisture exhibited slower diffusion rates. In the single-component model, the uniform distribution of the overall moisture diffusion coefficient results in a stratified decrease in moisture content from the exterior to the interior. The highest moisture concentration shifting toward the

geometric center of the kernel. In contrast, the multi-component model revealed a more complex moisture distribution pattern, with the highest moisture content localized away from the geometric center, particularly in the germ and portions of the soft endosperm. This highlights the influence of component-specific diffusion properties on moisture migration dynamics.

To validate the simulated moisture distribution results, proton density images of maize kernels were acquired using MRI throughout the drying process. These MRI measurements provided direct, spatially resolved data on moisture distribution, enabling a quantitative comparison with the simulation predictions. Fig. 8 presents MRI images of maize kernels dried at different temperatures. The moisture levels are identified by different color of the MRI images. As the drying process progresses, the red regions gradually diminish, while yellow and light blue regions expand, indicating the migration of internal moisture toward the surface. Simultaneously, surface moisture increases and evaporates outward, resulting in uneven moisture distribution within the kernels. After 180 min of drying at 60 °C, the germ retains relatively high moisture levels, consistent with results obtained from the multi-component model. A comparison of MRI images across different drying temperatures reveals that higher temperatures accelerate the

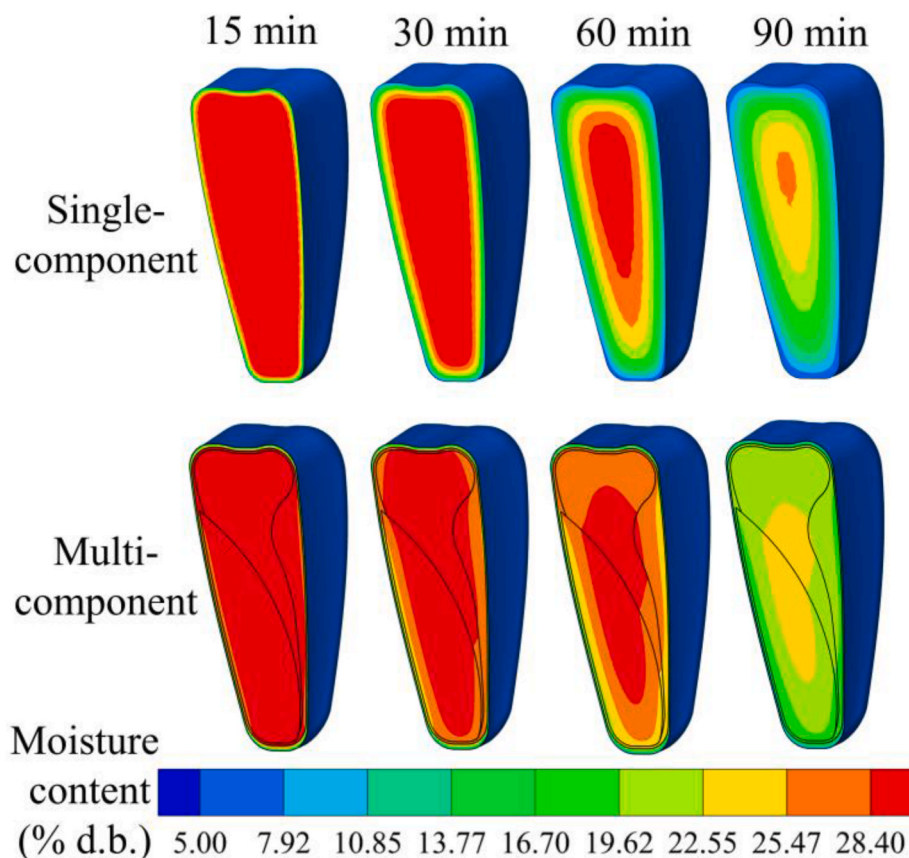


Fig. 7. Simulation moisture distribution in maize kernels based on two assumptions at different times under 60 °C drying temperature.

reduction in red regions. By the late stages of drying at 70 °C and 80 °C, the red regions disappear entirely. Increasing the drying temperature significantly promotes moisture loss. When comparing the simulated and experimental moisture distribution at 60 °C, the multi-component model demonstrates significantly higher accuracy in predicting moisture transfer within maize kernels. This improved performance is particularly evident in capturing the heterogeneous moisture distribution patterns observed during the drying process.

Fig. 9 illustrates the moisture content variation along the length, width, and thickness directions during the drying process at 60 °C, as predicted by the multi-component model. The total drying time was 105 min, with moisture distribution recorded at 15-min intervals. The results revealed significant non-uniform moisture distribution patterns. During the initial drying phase, moisture content decreased rapidly, followed by progressively smaller variations at each 15-min intervals. This trend was consistent across all three spatial directions. Moisture migration occurred preferentially along shorter paths from the higher-moisture regions to lower-moisture regions. The moisture content decreased radially from the interior to the surface, with the outer layers approaching equilibrium moisture content. As drying progressed, moisture from the germ migrated toward the endosperm, leading to a localized increase in moisture content in the endosperm regions adjacent to the germ. Notably, the germ near the surface exhibited slower moisture loss compared to the germ near the endosperm. This behavior can be attributed to the low diffusivity of the seed coat, which acts as a barrier to moisture transfer. This finding aligns with previous studies demonstrating the resistive role of seed coats in the drying process of various agricultural products (Di et al., 2000; Wang et al., 2017).

4.5. Temperature variation during maize kernel drying

Hot air temperature is a critical factor influencing both drying

efficiency and product quality. To investigate the temperature variation during maize kernel drying, a drying temperature of 60 °C was selected, with an assumption of an initial uniform temperature distribution. Fig. 10 illustrates the temperature distribution within the maize kernels at different times, as predicted by the single-component and multi-component models. As drying progressed, a distinct temperature gradient developed within the maize kernels. Within the first minute of drying, the interior temperature remained relatively low, while the outer regions heated rapidly, reaching approximately 55 °C at the surface. This pattern reflects the characteristic outward-to-inward heating mechanism of the drying process. During the initial drying phase, a pronounced temperature gradient was observed at the surface, while the internal gradient remained relatively small, with heat transfer occurring progressively from the exterior to the interior. As drying continued, the internal temperature gradually increased. After 4 min of drying, the single-component model predicted near-uniform temperature distribution, whereas the multi-component model maintained a detectable temperature gradient, albeit with a narrowing difference between the center and the surface. By 6 min, both models achieved the target drying temperature of 60 °C, with a negligible temperature difference of approximately 0.1 °C, indicating stabilization of the temperature field. The observed temperature dynamics are influenced by the moisture-dependent thermal properties of the kernels, including thermal conductivity and specific heat capacity. The non-uniform moisture distribution resulting from moisture transfer directly affects the temperature field distribution in both models, as demonstrated in previous studies (Zhang et al., 2013).

5. Conclusions

Low-field nuclear magnetic resonance (LF-NMR) technology was employed to investigate the moisture diffusion characteristics of maize

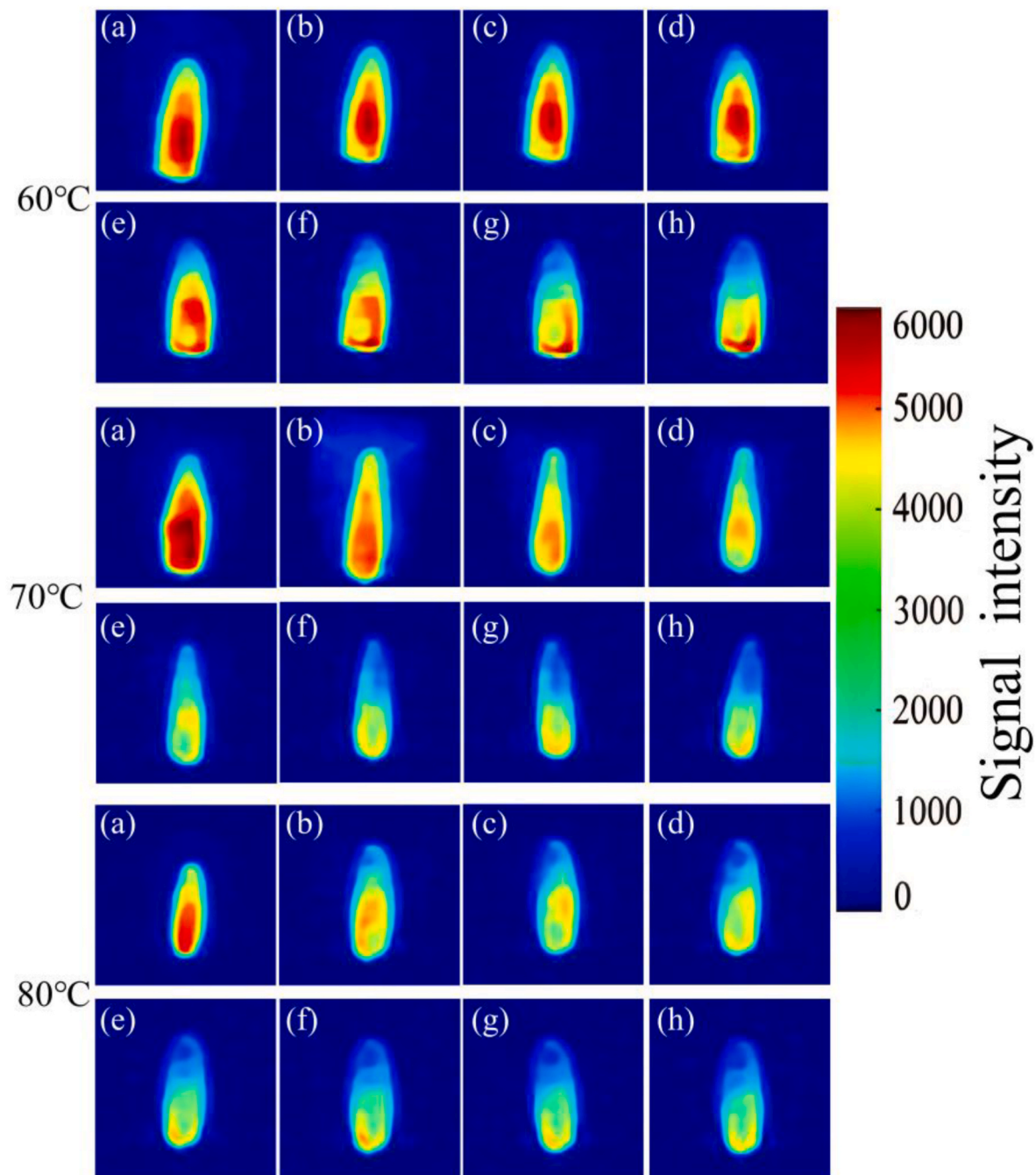


Fig. 8. MRI images of maize kernels during drying at different temperatures every 30 min.

kernels during hot air drying. A particle-scale model for maize drying was developed, accounting for the uniform moisture diffusion coefficients in single-component systems and the heterogeneity of multi-component systems. Experimental and simulation validations were conducted at various drying temperatures, and the main conclusions are as follows.

- (1) Changes in the transverse relaxation time (T_2) peak area during drying correlate strongly with variations in moisture content. As drying temperature increases, the T_{22} intensity peak and transverse relaxation time exhibit a pronounced leftward shift, indicating a significant reduction in water molecules mobility within the kernels. The binding strength of different moisture forms increases with decreasing moisture ratio but remains independent

of drying temperature. The moisture migration rate is influenced by both temperature and the relative proportions of the four distinct moisture forms present in the maize kernels.

- (2) The particle-scale geometric model of the maize kernel, combined with the drying model, accurately captures moisture content changes observed during hot air drying experiments. The multi-component model demonstrates high accuracy, with a maximum error of less than 4.96 % compared to experimental values. Component heterogeneity governs the non-uniform distribution of temperature and moisture content within the kernels during drying, with the germ exhibiting higher moisture content and lower temperature compared to the endosperm.
- (3) As drying progresses, moisture migrates from the germ to the endosperm, resulting in localized moisture accumulation in the

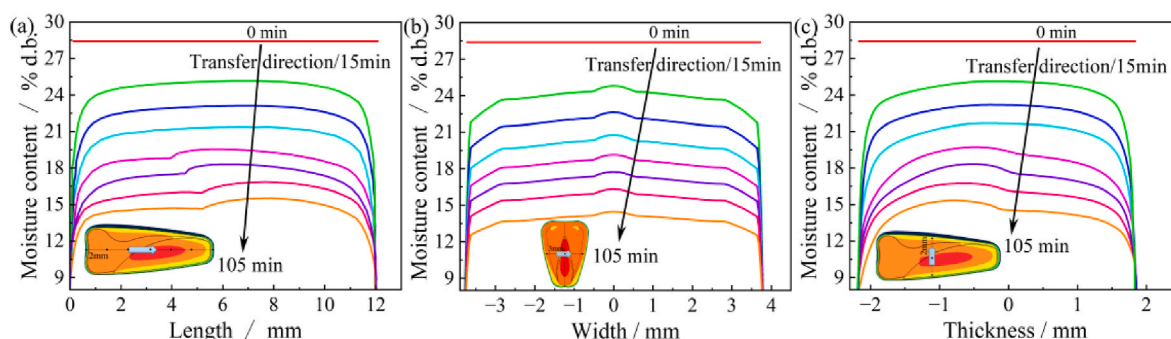


Fig. 9. Moisture content of maize kernels at different spatial locations in multi-component modeling under 60 °C drying temperature, (a) length, (b) width, (c) thickness.

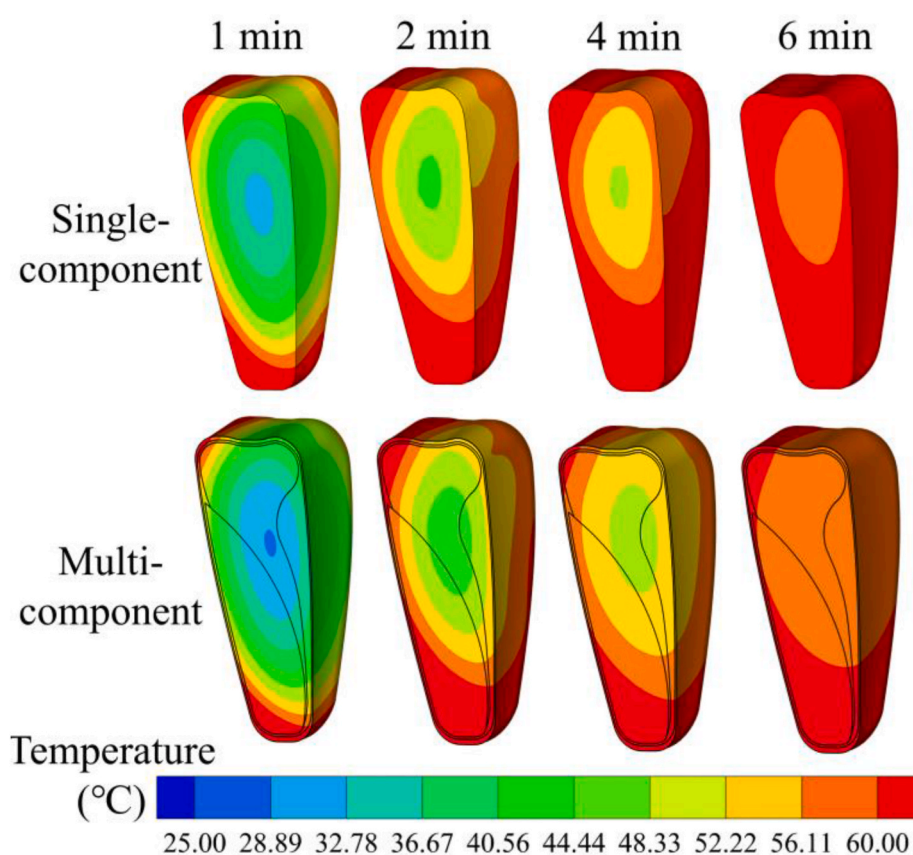


Fig. 10. Simulation temperature distribution in maize kernels based two assumptions at different times.

endosperm regions adjacent to the germ. This phenomenon delays germ drying, while the endosperm continues to dominate the overall drying process. The germ near the surface exhibits slower moisture loss compared to the germ near the endosperm, primarily due to the resistance imposed by the maize kernel's outer skin.

CRediT authorship contribution statement

Guixiang Chen: Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Wenlei Liu:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Deqian Zheng:** Writing – review & editing, Supervision, Funding acquisition. **Jun Wang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis. **Hao Zhang:** Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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